How Does Subway Proximity Relate to Advertised Rent in New York City?

A descriptive analysis of May 2026 rental listings

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Introduction

For New York City renters, subway access is part of the price of a home. A nearby station may shorten commutes and widen access to jobs, schools, services, and social life. A longer walk may be acceptable if it accompanies lower rent. Renters, landlords, and housing researchers therefore have reason to understand not whether transit access *causes* rent, but how strongly rents and access are associated in the advertised market.

The relationship is not self-evident. Listings near stations are concentrated in boroughs and neighborhoods with different housing stocks, while rents also vary with apartment size. A citywide average may combine a transit-access pattern with broad geographic and compositional differences. We ask the following research question. **Among New York City rental listings in May 2026, what is the descriptive relationship between advertised monthly rent and straight-line distance to the nearest subway station, and how does that relationship change after accounting for apartment size and borough?**

Data, Wrangling, and Operationalization

Rental data come from the FirstMover Open Data Project, which records a listing when it first appears on the market. Each row is an advertised listing, not a completed lease. Asking rent does not incorporate later price or status changes. Station names and centroid coordinates come from the MTA Subway Stations dataset.

The pipeline retains the five boroughs, removes 128 repeated URLs and 14 further repeated units, and computes Haversine distance from each listing coordinate to every station centroid. The closest centroid defines nearest-station distance. We reserved 30% of 20,385 geocoded listings for exploration and locked cleaning, transformations, and model selection before examining the 70% confirmation split. After deduplication, 20,243 listings remained. The final complete-case confirmation sample contains **3,101** listings. The reduction is driven mainly by missing square footage.

Advertised monthly rent operationalizes the price visible to renters but omits concessions, negotiation, total housing cost, and lease quality. Subway access is operationalized as straight-line meters to a station centroid, not walking distance or time to an entrance. Median rent in the analysis sample is **$4,634**. Its IQR is $3,495 to $6,500. Median station distance is **311 m**. Its IQR is 187 to 493 meters.

Visual Description

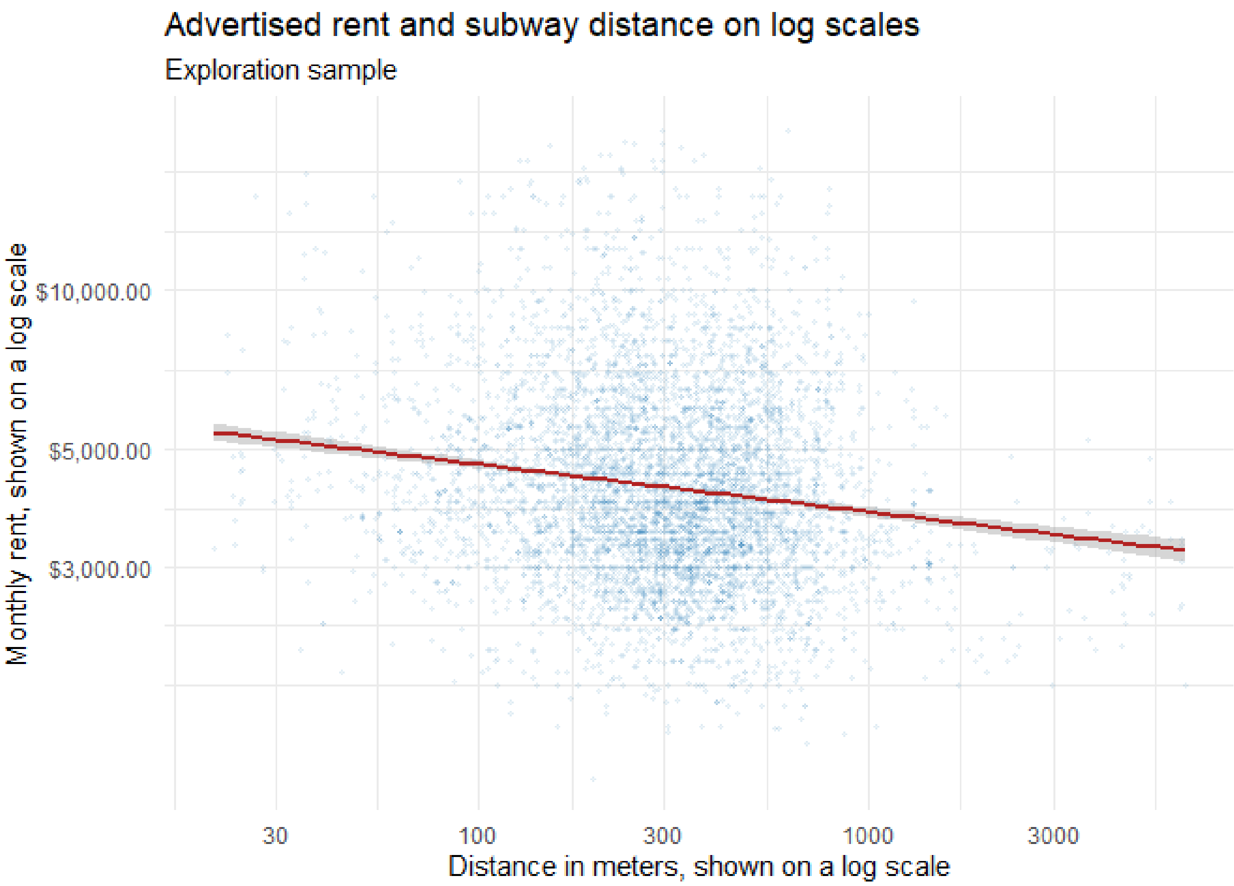


Figure 1 Advertised rent and subway distance on logarithmic axes in the exploration sample. The red line and confidence band show the exploratory linear fit from the supplied EDA Rmd.

The attached EDA figure shows a modest negative relationship on logarithmic axes. Advertised rents tend to be lower at greater subway distances, although listings at similar distances still span a wide range. The red line is the exploratory linear fit used to assess the relationship's shape.

The more even spread on logarithmic axes supported comparing transformed specifications during exploration. The broad vertical variation is substantive. Subway proximity is only one component of a larger bundle that includes location, apartment size, building quality, amenities, and transit service. The final `m3_loglog` results are reported in the regression table on the following page.

Models and Results

The simplest credible model summarizes the unconditional association in dollars per additional 100 meters.

$$\text{Rent}_i = \beta_0 + \beta_1(\text{Distance}_i/100) + \epsilon_i.$$

Exploration showed skew, curvature, and increasing residual variance in levels. Among the tested transformations, `m3_loglog` had the strongest fit and better residual behavior. We therefore locked the following model before confirmation.

$$\log(\text{Rent}_i) = \beta_0 + \beta_1 \log(\text{Distance}_i) + \beta_2\text{Bedrooms}_i + \beta_3\text{Bathrooms}_i + \beta_4\text{Sqft}_i + \boldsymbol{\gamma}'\text{Borough}_i + \epsilon_i.$$

During exploration, we also replaced borough with neighborhood indicators. This created 139 neighborhood groups in a much smaller complete-case sample. Many groups contained very few listings, which made the specification parameter-heavy and the individual estimates unstable. We therefore did not select the neighborhood model. The final `m3_loglog` specification uses borough indicators as the more stable geographic adjustment.

Table 1 Confirmation-sample estimates with HC1 robust standard errors in parentheses. Bronx is the reference borough.

	Simple model	Final `m3_loglog`
Intercept	5914.92 (66.59)	7.7473 (0.0614)
Distance	-86.94 (5.89)	-0.0714 (0.0064)
Bedrooms	–	0.0403 (0.0191)
Bathrooms	–	0.3082 (0.0320)
Square feet	–	0.00028 (0.00009)
Brooklyn	–	0.2997 (0.0353)
Manhattan	–	0.6367 (0.0346)
Queens	–	0.1018 (0.0352)
Staten Island	–	-0.3134 (0.0695)
R^2	0.030	0.616
Adjusted R^2	0.030	0.615
Observations	3,101	3,101

The borough coefficients compare otherwise similar listings with the omitted Bronx reference category. They are broad geographic adjustments rather than focal or causal estimates. The Staten Island contrast is especially uncertain because very few complete-case listings are represented.

In the simple model, an additional 100 meters is associated with **\$86.94 lower monthly rent** on average. Its R^2 of 0.030 shows that distance alone captures little listing-level variation.

In the final model, the distance elasticity is **-0.07137** (HC1 SE 0.00642). A two-sided robust coefficient test evaluates the null hypothesis $H_0 = \beta_{\log(\text{distance})} = 0$. It gives $t = -11.13$**,** $p = 3.22e - 28$. A 10% greater distance is associated with approximately **0.71% lower advertised rent**, holding bedrooms, bathrooms, square footage, and borough constant. Doubling distance corresponds to approximately **4.8% lower predicted rent**. The final model explains 61.6% of variation in log rent among complete-case confirmation listings, substantially more than the simple model. These estimates are statistically precise descriptions, not causal effects.

Assumptions, Limitations, and Interpretation

Functional form and large-sample inference. Logging rent and distance reduced curvature observed during exploration. The appendix diagnostic has no dominant parabolic pattern, but structured dispersion and outliers remain. With 3,101 confirmation observations, coefficient inference can rely on large-sample approximations despite non-normal residuals. HC1 standard errors address unequal variance, but the large sample does not correct misspecification, omitted variables, measurement error, influential outliers, or dependence among related listings.

Independence. Deduplication reduces repeated advertisements, but units in the same building or neighborhood may share unobserved conditions. Listing-level errors may therefore be correlated, making HC1 uncertainty estimates optimistic. Future work should consider building or neighborhood clustering where identifiers and group sizes permit it.

Measurement and selection. Centroid distance is measured with error relative to walking time or the nearest entrance. Square footage is missing for more than three quarters of all listings, so the complete-case sample may not represent the full advertised market. The model also omits neighborhood, building quality, amenities, transit lines and frequency, and listing-selection mechanisms. Borough indicators capture only broad geography.

Interpretive scope. The coefficient describes conditional differences between observed listings. Omitted location preferences and housing characteristics prevent a causal interpretation. "Holding variables constant" describes the regression comparison, not a feasible intervention that moves an otherwise identical apartment.

Conclusion

The May 2026 market contains a modest subway-access gradient after accounting for borough, bedrooms, bathrooms, and square footage. Greater distance is associated with lower advertised rent, but distance alone explains little of the market. Subway access is one visible component of a much larger spatial and housing-quality bundle. For renters, the fitted gradient is best understood as a market-level tradeoff rather than a fixed price per block. Future research could prespecify and validate neighborhood-level models, measure walking distance to station entrances, and examine whether the gradient differs across boroughs or apartment types.

Appendix

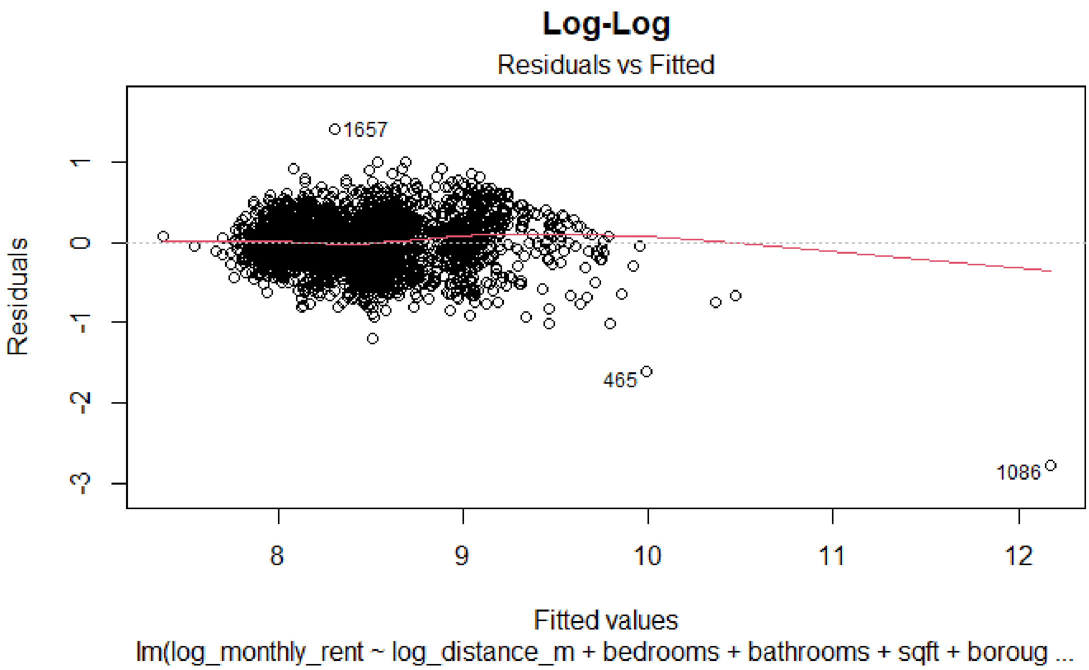
Data Access and Reproducibility

Rental listings come from the FirstMover Open Data Project, May 2026. Subway locations come from the MTA Subway Stations dataset. The complete pipeline is available in the project GitHub repository. It downloads, cleans, deduplicates, calculates Haversine distances, assigns the nearest station, splits with seed 203, and derives log variables.

Specifications Tried During Exploration

1. **Rent on distance only.** We retained this as the unconditional summary and test of no overall relationship.
2. **Levels with apartment controls and borough.** Fit improved, but skew and residual curvature remained.
3. **Log X or log Y alternatives.** Logging rent improved residual behavior. Logging both focal variables produced the strongest exploratory fit.
4. **Final `m3_loglog`.** This model uses log rent, log distance, bedrooms, bathrooms, square footage, and borough. We selected it before confirmation.
5. **Neighborhood indicators.** We tried this only during exploration. It produced 139 groups in a smaller complete-case sample. Many groups were sparse, so the model was too parameter-heavy for the final specification.

Final-Model Diagnostic



The residual cloud has no single dominant curve, supporting the log-log specification as a compact description. Remaining spread and clusters are consistent with heteroskedasticity and omitted geographic or building features.